



United International University
Department of CSE
CSE 3422: Software Engineering Laboratory
Lab Sheet

Experiment No. 1: Implementing CRUD Operations

1. Introduction

In computer programming, Create, Read, Update, and Delete (CRUD) are the four basic operations of persistent storage. Data can be put in a location/area of a storage mechanism.

- The fundamental feature of a storage location is that its content is both readable and updatable.
- Before a storage location can be read or updated it needs to be created; that is allocated and initialized with content.
- At some later point the storage location may need to be destructed; that is finalized and de-allocated.

Together these four operations make up the basic operations of storage management known as CRUD: Create, Read, Update, and Delete.

2. Lab Equipment/Required Software

- a. XAMPP (<https://www.apachefriends.org/download.html>)

3. Objective

- a. Get familiar with the XAMPP and the PHP.
- b. Learn how to use PHPMysqlAdmin.

4. Workflow

- a. Download necessary software if required (Section - 2).
- b. Download the sample CRUD compressed file from Google Drive
- c. Copy the ‘crud’ folder to the XAMPP htdocs directory.
- d. Create sample database table using PHPMysqlAdmin.
- e. Inspect the PHP files.
- f. Deploy the system locally using XAMPP and use the features in localhost.

5. Hand's on Practice

- a. Create a database with the following specifications:

Database Name: Fall2025_CW	
Table	Columns
Teacher	teacher_id, name, course_id
Course	course_id, title

- b. Create the following APIs:
 - i. Read Teachers
 - ii. Read Courses
 - iii. Assign Teacher To Course
 - iv. A home page to show all the links

6. Homework

- a. Create a database with the following specifications:

Database Name: SchoolMgmt
TABLE student: id, dept, name, nid, birth, address, etc. TABLE teacher: id, dept, name, nid, birth, address, etc. TABLE course: id, dept, title, credit, syllabus, etc. TABLE payment: payment_id, student_id, amount, date, etc.

- b. Create the following APIs:

Student createStudent(id, dept, name, nid, birth, address) readStudent(dept, batch) updateStudent(id, dept, name, nid, birth, address) deleteStudent(id)
Teacher createTeacher(id, dept, name, nid, birth, address) readTeacher(dept) updateTeacher(id, dept, name, nid, birth, address) deleteTeacher(id)
Course createCourse(id, dept, title, credit, syllabus) readCourse(dept, semester) updateCourse(id, dept, title, credit, syllabus) deleteCourse(id)
Payment createPayment(payment_id, student_id, amount, date) readPayment(student_id) updatePayment(payment_id, student_id, amount, date) deletePayment(payment_id)
Cross-Module API addCourseToTeacher(course_id, teacher_id) createCourseToSemester(course_id, dept, semester) removeCourseFromSemester(course_id, semester) addCourseToTeacher(course_id, teacher_id) removeCourseFromTeacher(course_id, teacher_id)